

FORMULAS III

CHAPTER 6: 2^k -DESIGN

2^2
-Factor
Design

Factor		Treatment Combination	Replicate		
A	B		I	II	III
–	–	A low, B low	28	25	27
+	–	A high, B low	36	32	32
–	+	A low, B high	18	19	23
+	+	A high, B high	31	30	29

Main effects and 2nd order interaction effects

The effect of A at the low level of B is $[a - (1)]/n$ and at the high level of B it is $[ab - b]/n$. Averaging these quantities yields the **main effect** of A . Applying this principle also onto B and AB gives

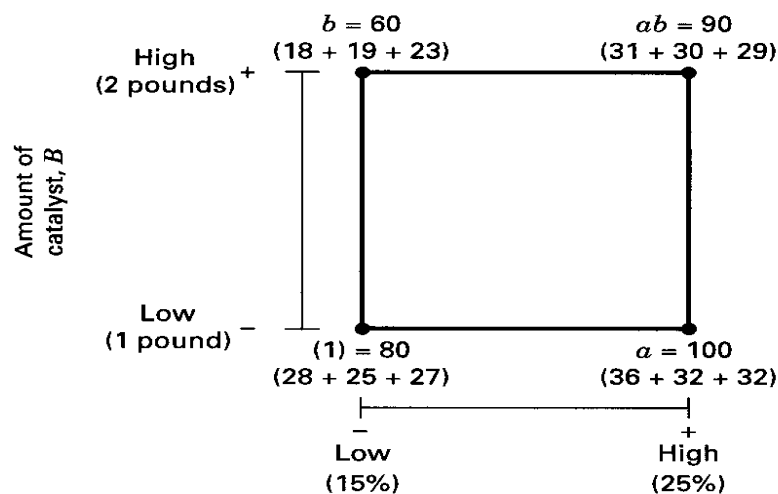
$$A = \frac{1}{2n} \{[ab - b] + [a - (1)]\} = \frac{1}{2n} \{[ab + a] - [b + (1)]\}$$

$$B = \frac{1}{2n} \{[ab - a] + [b - (1)]\} = \frac{1}{2n} \{[ab + b] - [a + (1)]\}$$

$$AB = \frac{1}{2n} \{[ab - b] - [a - (1)]\} = \frac{1}{2n} \{[ab + (1)] - [a + b]\}$$

$$AB = \frac{1}{2n} \{[ab - b] - [a - (1)]\}$$

We define the **interaction effect** AB as the average difference between the effect of A at the high level of B and the effect of A at the low level of B . Thus,



$$AB = \frac{1}{4n} [ab - a - b + (1) + abc - bc - ac + c]$$

Can write effect of A as

$$\begin{aligned} A &= \bar{y}_{A^+} - \bar{y}_{A^-} \\ &= \frac{a + ab + ac + abc}{4n} - \frac{(1) + b + c + bc}{4n} \end{aligned}$$

Can write effect of AB as,

$$AB = \frac{abc + ab + c + (1)}{4n} - \frac{bc + b + ac + a}{4n}$$

<i>B</i>	Average A Effect
High (+)	$\frac{[(abc - bc) + (ab - b)]}{2n}$
Low (-)	$\frac{\{(ac - c) + [a - (1)]\}}{2n}$

General:

$$\begin{aligned} A &= \bar{y}_{A^+} - \bar{y}_{A^-} \\ B &= \bar{y}_{B^+} - \bar{y}_{B^-} \\ C &= \bar{y}_{C^+} - \bar{y}_{C^-} \\ \text{etc, etc, ...} \end{aligned}$$

**2⁴
Factor
Desgin**

Table 6-10 Pilot Plant Filtration Rate Experiment

Run Number	Factor				Run Label	Filtration Rate (gal/h)
	A	B	C	D		
1	-	-	-	-	(1)	45
2	+	-	-	-	a	71
3	-	+	-	-	b	48
4	+	+	-	-	ab	65
5	-	-	+	-	c	68
6	+	-	+	-	ac	60
7	-	+	+	-	bc	80
8	+	+	+	-	abc	65
9	-	-	-	+	d	43
10	+	-	-	+	ad	100
11	-	+	-	+	bd	45
12	+	+	-	+	abd	104
13	-	-	+	+	cd	75
14	+	-	+	+	acd	86
15	-	+	+	+	bcd	70
16	+	+	+	+	abcd	96

2^k Factor Design	The methods of analysis that we have presented thus far may be generalized to the case of a 2^k factorial design, that is, a design with k factors each at two levels. The statistical model for a 2^k design would include k main effects, $\binom{k}{2}$ two-factor interactions, $\binom{k}{3}$ three-factor interactions, . . . , and one k-factor interaction. That is, for a 2^k design the complete model would contain $2^k - 1$ effects. The notation introduced earlier for treatment combinations is also used here. For example, in a 2^5 design abd denotes the treatment combination with factors A, B, and D at the high level and factors C and E at the low level.
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CHAPTER 7: BLOCKING WITH 2^k FACTOR DESIGN

2^k-Factor design 2²	Single replicate (AB is confounded with blocks) Block 1 (1) ab Block 2 a b																																																																																									
2³	Multiple replicates (replicates forming blocks)(suppose n=3) Block 1 (1) = 28 a = 36 b = 18 ab = 31 Block 2 (1) = 25 a = 32 b = 19 ab = 30 Block 3 (1) = 27 a = 32 b = 23 ab = 29																																																																																									
2³ Design	■ TABLE 6.3 Algebraic Signs for Calculating Effects in the 2³ Design <table><tr><th rowspan="2">Treatment Combination</th><th colspan="8">Factorial Effect</th></tr><tr><th>I</th><th>A</th><th>B</th><th>AB</th><th>C</th><th>AC</th><th>BC</th><th>ABC</th></tr><tr><td>(1)</td><td>+</td><td>−</td><td>−</td><td>+</td><td>−</td><td>+</td><td>+</td><td>−</td></tr><tr><td>a</td><td>+</td><td>+</td><td>−</td><td>−</td><td>−</td><td>−</td><td>+</td><td>+</td></tr><tr><td>b</td><td>+</td><td>−</td><td>+</td><td>−</td><td>−</td><td>+</td><td>−</td><td>+</td></tr><tr><td>ab</td><td>+</td><td>+</td><td>+</td><td>+</td><td>−</td><td>−</td><td>−</td><td>−</td></tr><tr><td>c</td><td>+</td><td>−</td><td>−</td><td>+</td><td>+</td><td>−</td><td>−</td><td>+</td></tr><tr><td>ac</td><td>+</td><td>+</td><td>−</td><td>−</td><td>+</td><td>+</td><td>−</td><td>−</td></tr><tr><td>bc</td><td>+</td><td>−</td><td>+</td><td>−</td><td>+</td><td>−</td><td>+</td><td>−</td></tr><tr><td>abc</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr></table> Single replicate (blocking using the signs of interaction factor ABC)	Treatment Combination	Factorial Effect								I	A	B	AB	C	AC	BC	ABC	(1)	+	−	−	+	−	+	+	−	a	+	+	−	−	−	−	+	+	b	+	−	+	−	−	+	−	+	ab	+	+	+	+	−	−	−	−	c	+	−	−	+	+	−	−	+	ac	+	+	−	−	+	+	−	−	bc	+	−	+	−	+	−	+	−	abc	+	+	+	+	+	+	+	+
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bc	+	−	+	−	+	−	+	−																																																																																		
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	Block 1 (1) <i>ab</i> <i>ac</i> <i>bc</i> Block 2 <i>a</i> <i>b</i> <i>c</i> <i>abc</i> Multiple Replicates (suppose n=4) : 2 Blocks B1, B2 from each replicate group Replicate I Block 1 (1) <i>ac</i> <i>ab</i> <i>bc</i> Block 2 <i>abc</i> <i>a</i> <i>b</i> <i>c</i> Replicate II Block 1 (1) <i>ac</i> <i>ab</i> <i>bc</i> Block 2 <i>abc</i> <i>a</i> <i>b</i> <i>c</i> Replicate III Block 1 (1) <i>ac</i> <i>ab</i> <i>bc</i> Block 2 <i>abc</i> <i>a</i> <i>b</i> <i>c</i> Replicate IV Block 1 (1) <i>ac</i> <i>ab</i> <i>bc</i> Block 2 <i>abc</i> <i>a</i> <i>b</i> <i>c</i>
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CHAPTER 8: FRACTIONAL FACTOR DESIGN [One-Half and One-Quarter Fractional]

One-Half Factor Design 2^{k-1}	• Notation: because the design has 2^{k-1} runs, it's referred to as a 2^{k-1} • Consider a really simple case, the 2^{3-1}
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	<table><tr><th colspan="9">Table 8-1 Plus and Minus Signs for the 2^3 Factorial Design</th></tr><tr><th rowspan="2">Treatment Combination</th><th colspan="8">Factorial Effect</th></tr><tr><th>I</th><th>A</th><th>B</th><th>C</th><th>AB</th><th>AC</th><th>BC</th><th>ABC</th></tr><tr><td>a</td><td>+</td><td>+</td><td>-</td><td>-</td><td>-</td><td>-</td><td>+</td><td>+</td></tr><tr><td>b</td><td>+</td><td>-</td><td>+</td><td>-</td><td>-</td><td>+</td><td>-</td><td>+</td></tr><tr><td>c</td><td>+</td><td>-</td><td>-</td><td>+</td><td>+</td><td>-</td><td>-</td><td>+</td></tr><tr><td>abc</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td></tr><tr><td>ab</td><td>+</td><td>+</td><td>+</td><td>-</td><td>+</td><td>-</td><td>-</td><td>-</td></tr><tr><td>ac</td><td>+</td><td>+</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>-</td></tr><tr><td>bc</td><td>+</td><td>-</td><td>+</td><td>+</td><td>-</td><td>-</td><td>+</td><td>-</td></tr><tr><td>(1)</td><td>+</td><td>-</td><td>-</td><td>-</td><td>+</td><td>+</td><td>+</td><td>-</td></tr></table>	Table 8-1 Plus and Minus Signs for the 2^3 Factorial Design									Treatment Combination	Factorial Effect								I	A	B	C	AB	AC	BC	ABC	a	+	+	-	-	-	-	+	+	b	+	-	+	-	-	+	-	+	c	+	-	-	+	+	-	-	+	abc	+	+	+	+	+	+	+	+	ab	+	+	+	-	+	-	-	-	ac	+	+	-	+	-	+	-	-	bc	+	-	+	+	-	-	+	-	(1)	+	-	-	-	+	+	+	-
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	Defining relation: $I = ABC$ (top) $I = -ABC$ (bottom)																																																																																																		
Aliases-Resolution	Aliases can be found from the defining relation $I = ABC$ by multiplication: $AI = A(ABC) = A^2BC = BC$ $BI = B(ABC) = AC$ $CI = C(ABC) = AB$ “Main effects aliased with 2 nd order interaction” Resolution III (1 + 2 = 3)																																																																																																		
Building One Half - Design	Can build from 2^2 up (general 2^k design up)																																																																																																		
Build 2^{3-1} From Full 2^2	<table><tr><th colspan="9">Table 8-2 The Two One-Half Fractions of the 2^3 Design</th></tr><tr><th rowspan="2">Run</th><th colspan="2">Full 2^2 Factorial (Basic Design)</th><th colspan="3">$2^{3-1}_{III}, I = ABC$</th><th colspan="3">$2^{3-1}_{III}, I = -ABC$</th></tr><tr><th>A</th><th>B</th><th>A</th><th>B</th><th>$C = AB$</th><th>A</th><th>B</th><th>$C = -AB$</th></tr><tr><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>+</td><td>-</td><td>-</td><td>-</td></tr><tr><td>2</td><td>+</td><td>-</td><td>+</td><td>-</td><td>-</td><td>+</td><td>-</td><td>+</td></tr><tr><td>3</td><td>-</td><td>+</td><td>-</td><td>+</td><td>-</td><td>-</td><td>+</td><td>+</td></tr><tr><td>4</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>+</td><td>-</td></tr></table>	Table 8-2 The Two One-Half Fractions of the 2^3 Design									Run	Full 2^2 Factorial (Basic Design)		$2^{3-1}_{III}, I = ABC$			$2^{3-1}_{III}, I = -ABC$			A	B	A	B	$C = AB$	A	B	$C = -AB$	1	-	-	-	-	+	-	-	-	2	+	-	+	-	-	+	-	+	3	-	+	-	+	-	-	+	+	4	+	+	+	+	+	+	+	-																																				
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3	-	+	-	+	-	-	+	+																																																																																											
4	+	+	+	+	+	+	+	-																																																																																											
(need Defining Relation)	<table><tr><th colspan="7">Table 8-3 The 2^{4-1}_{IV} Design with the Defining Relation $I = ABCD$</th></tr><tr><th rowspan="2">Run</th><th colspan="3">Basic Design</th><th rowspan="2">$D = ABC$</th><th rowspan="2">Treatment Combination</th><th rowspan="2">Filtration Rate</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>(1)</td><td>45</td></tr><tr><td>2</td><td>+</td><td>-</td><td>-</td><td>+</td><td>ad</td><td>100</td></tr><tr><td>3</td><td>-</td><td>+</td><td>-</td><td>+</td><td>bd</td><td>45</td></tr><tr><td>4</td><td>+</td><td>+</td><td>-</td><td>-</td><td>ab</td><td>65</td></tr><tr><td>5</td><td>-</td><td>-</td><td>+</td><td>+</td><td>cd</td><td>75</td></tr><tr><td>6</td><td>+</td><td>-</td><td>+</td><td>-</td><td>ac</td><td>60</td></tr><tr><td>7</td><td>-</td><td>+</td><td>+</td><td>-</td><td>bc</td><td>80</td></tr><tr><td>8</td><td>+</td><td>+</td><td>+</td><td>+</td><td>$abcd$</td><td>96</td></tr></table>	Table 8-3 The 2^{4-1}_{IV} Design with the Defining Relation $I = ABCD$							Run	Basic Design			$D = ABC$	Treatment Combination	Filtration Rate	A	B	C	1	-	-	-	-	(1)	45	2	+	-	-	+	ad	100	3	-	+	-	+	bd	45	4	+	+	-	-	ab	65	5	-	-	+	+	cd	75	6	+	-	+	-	ac	60	7	-	+	+	-	bc	80	8	+	+	+	+	$abcd$	96																									
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One-Quarter Design 2^{k-2} Design Need 2 Generators	Table 8-9 Construction of the 2_{IV}^{6-2} Design with the Generators $I = ABCE$ and $I = BCDF$						
	Basic Design						
	Run	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E = ABC</i>	<i>F = BCD</i>
	1	–	–	–	–	–	–
	2	+	–	–	–	+	–
	3	–	+	–	–	+	+
	4	+	+	–	–	–	+
	5	–	–	+	–	+	+
	6	+	–	+	–	–	+
	7	–	+	+	–	–	–
	8	+	+	+	–	+	–
	9	–	–	–	+	–	+
	10	+	–	–	+	+	+
	11	–	+	–	+	+	–
	12	+	+	–	+	–	–
	13	–	–	+	+	+	–
	14	+	–	+	+	–	–
	15	–	+	+	+	–	+
	16	+	+	+	+	+	+

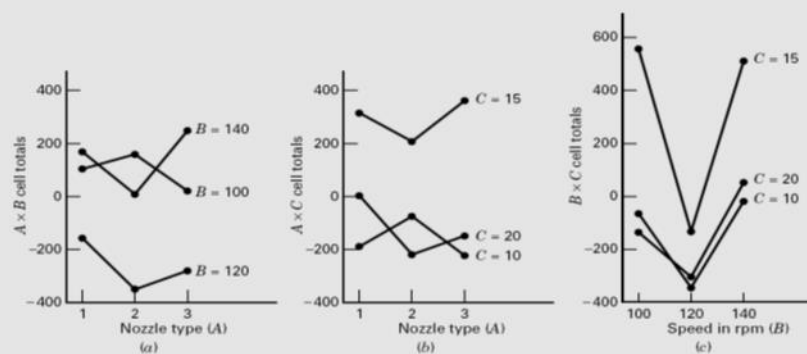
CHAPTER 9: 3^k FACTOR DESIGN

3^2 Design	In the 3^k system of designs, when the factors are quantitative, we often denote the low, intermediate, and high levels by -1, 0, and $+1$, respectively. This facilitates fitting a regression model relating the response to the factor levels. For example, consider the 3^2 design in Figure 9.1, and let x_1 represent factor <i>A</i> and x_2 represent factor <i>B</i>. A regression model relating the response y to x_1 and x_2 that is supported by this design is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \epsilon \quad (9.1)$ Notice that the addition of a third factor level allows the relationship between the response and design factors to be modeled as a quadratic.
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3³ Design

■ **TABLE 9.2**
Analysis of Variance for Syrup Loss Data

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P -Value
A, nozzle	993.77	2	496.89	1.17	0.3256
B, speed	61,190.33	2	30,595.17	71.74	<0.0001
C, pressure	69,105.33	2	34,552.67	81.01	<0.0001
AB	6,300.90	4	1,575.22	3.69	0.0160
AC	7,513.90	4	1,878.47	4.40	0.0072
BC	12,854.34	4	3,213.58	7.53	0.0003
ABC	4,628.76	8	578.60	1.36	0.2580
Error	11,515.50	27	426.50		
Total	174,102.83	53			



■ **FIGURE 9.4** Two-factor interactions for Example 9.1